

实验三：Simulink 的基本模块的应用及建模程序结构的方法

实验目的：

熟悉 Simulink 的工作环境，能够运用基本模块建立模型，并且进行仿真，观察仿真结果；掌握建模程序结构的两种方法：块图方法和运用 Embedded MATLAB function 块图的方法。

实验内容：

1. A transducer takes position and converts it to speed. The mathematical model for a transducer is an algebraic equation. Use blocks from the simulink library to implement a transducer.

The transducer is modeled with the following equation

$$y=3u+2$$

Where u represents position and y represents velocity.

- a) Create a Simulink model of the transducer. Use a Ramp block (with the default value of 1 for the **Slope**) from the Sources library to view the position (u). Use a Display block from the Sinks library to view the value of the velocity (y) signal. The display block shows you only the current value of the signal.
- b) Add a Signal Viewer to the velocity signal to see the signal values over the course of the entire simulation.
- c) Set up the model to use input data from the MATLAB workspace.
 - (I) Delete the Ramp block and replace it with an Inport block from the Sources library.
 - (II) Set up the data import parameters such that the following inputs are used.

time	position(u)
0.0	0.0
3.0	0.5
7.0	1.0
10.0	0.0

- d) Output the velocity signal to the MATLAB workspace and plot it using the plot function.

Hint:

Input: u —Position; output: y ---Velocity

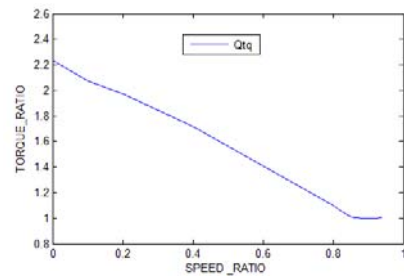
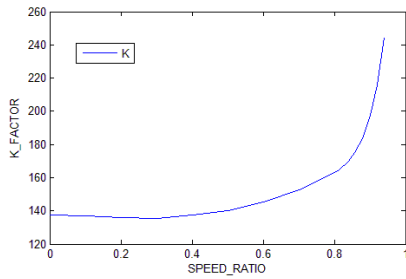
2. In a torque converter system, the impeller torque and turbine torque can be represented with the following equations.

$$T_{impeller} = (N_{engine} / K)^2$$

$$T_{turbine} = Q_{tq} T_{impeller}$$

$$SpeedRatio = N_{turbine} / N_{engine}$$

In the above equations, the variables Q_{tq} and K are known via the following empirical relationships.



Create the model using the equations and empirical relationships above.

Hint

Inputs: N_{engine} —Engine speed (N_{engine}) ;

$N_{turbine}$ —Turbine speed ($N_{turbine}$)

t —Time(t)

Output: $T_{turbine}$ —turbine torque

Parameters: $SpeedRatio$ ---Speed ratio reference for lookup table ($SPEED_RATIO$)

K — K factor as a function of speed ratio (K_FACTOR)

Q_{tq} —Torque ratio as a function of speed ratio ($TORQUE_RATIO$)

All the necessary parameters and input signals are loaded automatically into the MATLAB workspace from `torque_converter.mat` as a model preload function.

3. Diodes are commonly used to allow current to pass in only one direction. You can model the voltage/current relationship of a power diode using a lookup table.

The input (voltage)/output (current) relationship is known to be as follows:

Input Data Map (Volts)	Table Data (Amps)
-5	-0.4
-4	-0.2
-3	-0.1
-2	-0.05
-1	-0.01
0	0.0
1	0.6
2	1.5
3	3.2
4	4.0
5	4.1

Use the data above to define the parameters of the lookup table,

Test your power diode model with a Sine wave input (with amplitude of 5). Attach a single Signal Viewer to both the input and output of the diode so that signals appear on the same display.

Hint:

Input: V-Voltage(sine wave with amplitude 5)

Output: i —Current

Parameter: Input data map (volts) Table data (amps)

4. The Mars Pathfinder was launched outside the atmosphere of Mars with a parachute that opens automatically based on the altitude of the Pathfinder as well as its velocity. The parachute is deployed in the altitude range 6500-17000m. The mathematical model for the opening of the parachute is a logical equation.

Model the logic of the parachute two ways

- a) Using Simulink logical blocks
- b) Using the Embedded MATLAB function block.

The parachute logic is summarized in the following pseudo code:

```
deploy=0
if(altitude<=17000 m) and (vel<=503.8 m/s)
    deploy=1
if (altitude<=6500 m)
    deploy=1
```

Test data for your system inputs, altitude and velocity, is in `mars_para_data.mat` (load `mars_para_data`). Use inport blocks to pass altitude and velocity into the model (you can use the starter model `mars_para_start`). Simulate the model using a variable-step, discrete solver with a **Stop time** of 200. Determine the time that the parachute was deployed.

Hint: Duplicate Inport blocks to avoid crossing lines when sharing input signals.

Inputs:

altitude—Altitude of pathfinder(A l t)

vel—Velocity of pathfinder (V e l)

t—Time(t)

output: *deploy*—Boolean indicating that parachute deployed